

COGNIZANT

Mandatory Hands-on Exercises

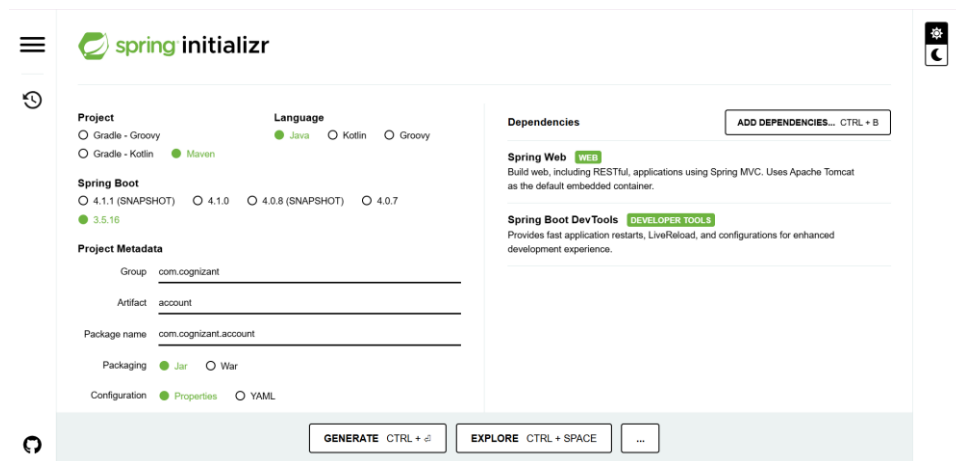
-Khushi Ahi

TOPIC: Microservices

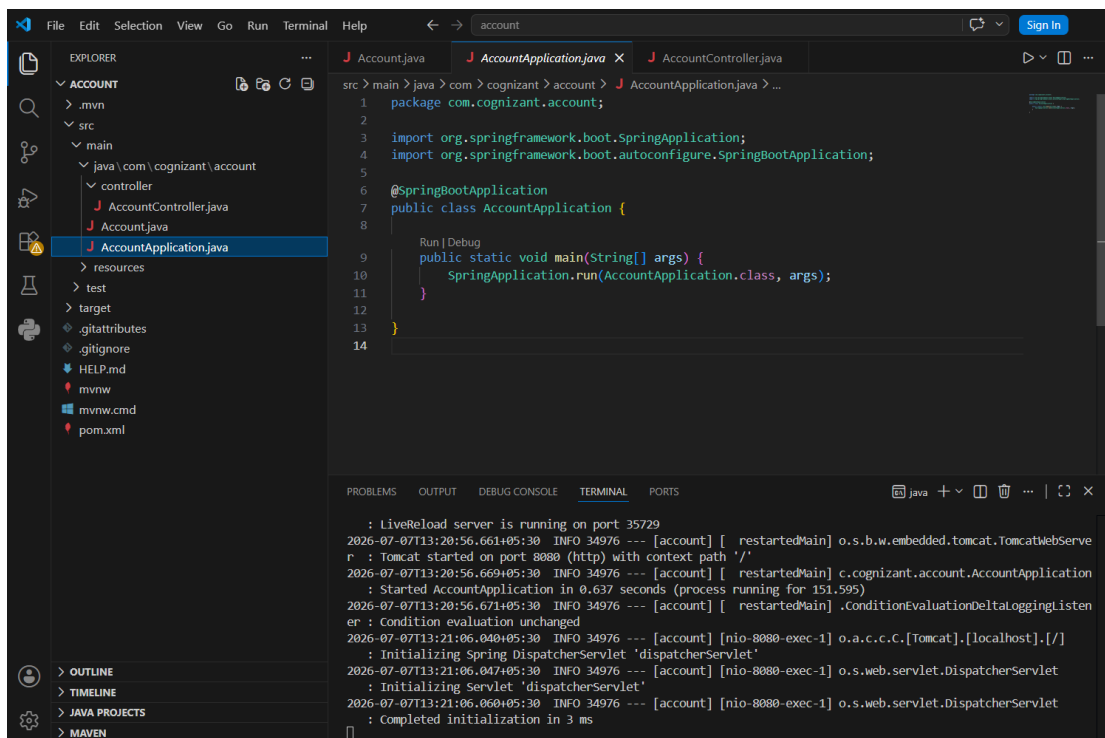
Exercise 1: Creating Microservices for account and loan

Account:

CODE



The image shows the Spring Initializr web interface. It is a form for generating a Spring project. The 'Project' section has 'Gradle - Groovy' selected. The 'Language' section has 'Java' selected. The 'Spring Boot' section has '4.1.1 (SNAPSHOT)' selected. The 'Project Metadata' section has 'Group' as 'com.cognizant', 'Artifact' as 'account', 'Package name' as 'com.cognizant.account', 'Packaging' as 'Jar', and 'Configuration' as 'Properties'. The 'Dependencies' section has 'Spring Web' and 'Spring Boot DevTools' selected. There are buttons for 'GENERATE', 'EXPLORE', and 'ADD DEPENDENCIES...'.



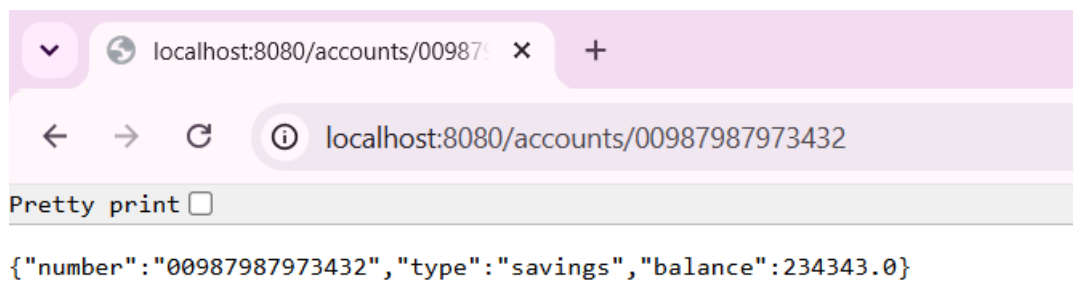
```
src > main > java > com > cognizant > account > AccountApplication.java > ...
1 package com.cognizant.account;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class AccountApplication {
8
9     Run | Debug
10     public static void main(String[] args) {
11         SpringApplication.run(AccountApplication.class, args);
12     }
13 }
14
```

The image shows a VS Code editor with the 'AccountApplication.java' file open. The code is a Spring Boot application. The terminal output shows the application running successfully on port 8080.

```
src > main > java > com > cognizant > account > controller > AccountController.java > AccountController
1 package com.cognizant.account.controller;
2
3 import com.cognizant.account.Account;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.PathVariable;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class AccountController {
10
11     @GetMapping("/accounts/{number}")
12     public Account getAccount(@PathVariable String number) {
13
14         return new Account(
15             number,
16             type: "savings",
17             balance: 234343
18         );
19     }
20 }
```

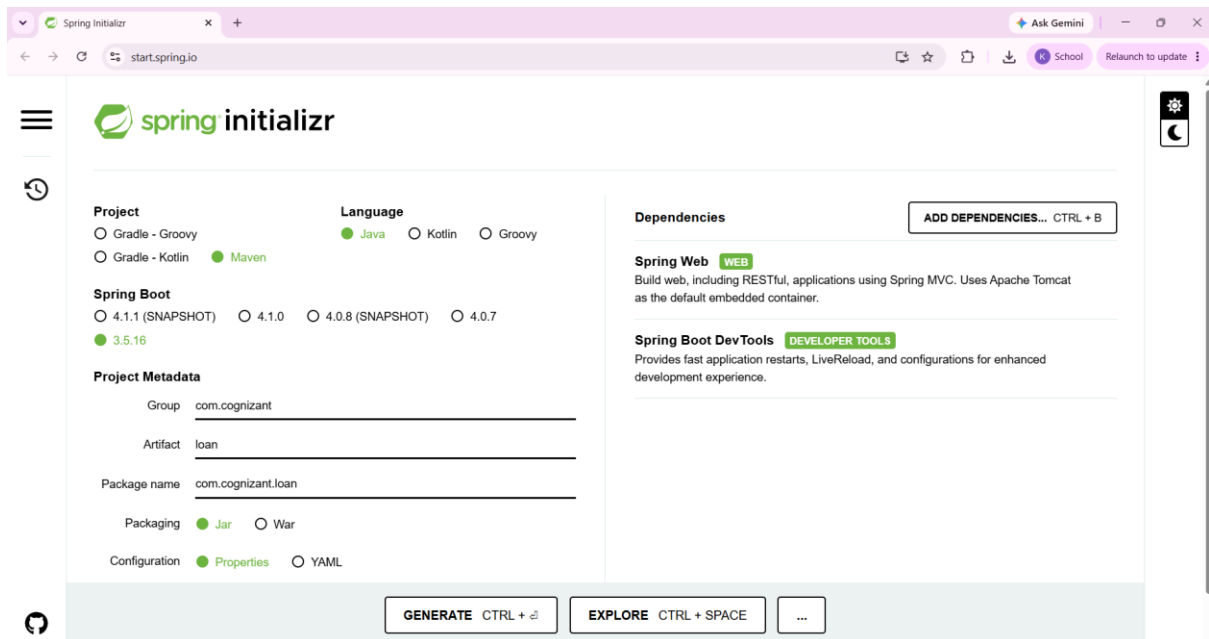
```
src > main > java > com > cognizant > account > Account.java > Account > getType()
1 package com.cognizant.account;
2 public class Account {
3
4     private String number;
5     private String type;
6     private double balance;
7
8     public Account() {
9     }
10
11     public Account(String number, String type, double balance) {
12         this.number = number;
13         this.type = type;
14         this.balance = balance;
15     }
16     public String getNumber() {
17         return number;
18     }
19     public void setNumber(String number) {
20         this.number = number;
21     }
22     public String getType() {
23         return type;
24     }
25     public void setType(String type) {
26         this.type = type;
27     }
28
29     public double getBalance() {
30         return balance;
31     }
32
33     public void setBalance(double balance) {
34         this.balance = balance;
35     }
36 }
```

OUTPUT



Loan:

CODE



The Spring Initializr web interface is shown in a browser window. The URL is start.spring.io. The page has a sidebar with a hamburger menu and a search icon. The main content area is divided into sections: Project, Language, Spring Boot, Project Metadata, and Dependencies. The Project section has radio buttons for Gradle - Groovy, Gradle - Kotlin, and Maven (selected). The Language section has radio buttons for Java (selected), Kotlin, and Groovy. The Spring Boot section has radio buttons for 4.1.1 (SNAPSHOT), 4.1.0, 4.0.8 (SNAPSHOT), and 4.0.7, with 3.5.16 selected. The Project Metadata section has input fields for Group (com.cognizant), Artifact (loan), and Package name (com.cognizant.loan). The Packaging section has radio buttons for Jar (selected) and War. The Configuration section has radio buttons for Properties (selected) and YAML. The Dependencies section has a button to ADD DEPENDENCIES... CTRL + B and lists Spring Web (WEB) and Spring Boot DevTools (DEVELOPER TOOLS). At the bottom, there are buttons for GENERATE CTRL + G, EXPLORE CTRL + SPACE, and a menu icon.

Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ Maven

Language

☒ Java ☐ Kotlin ☐ Groovy

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☐ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☒ 3.5.16

Project Metadata

Group

Artifact

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

Dependencies

ADD DEPENDENCIES... CTRL + B

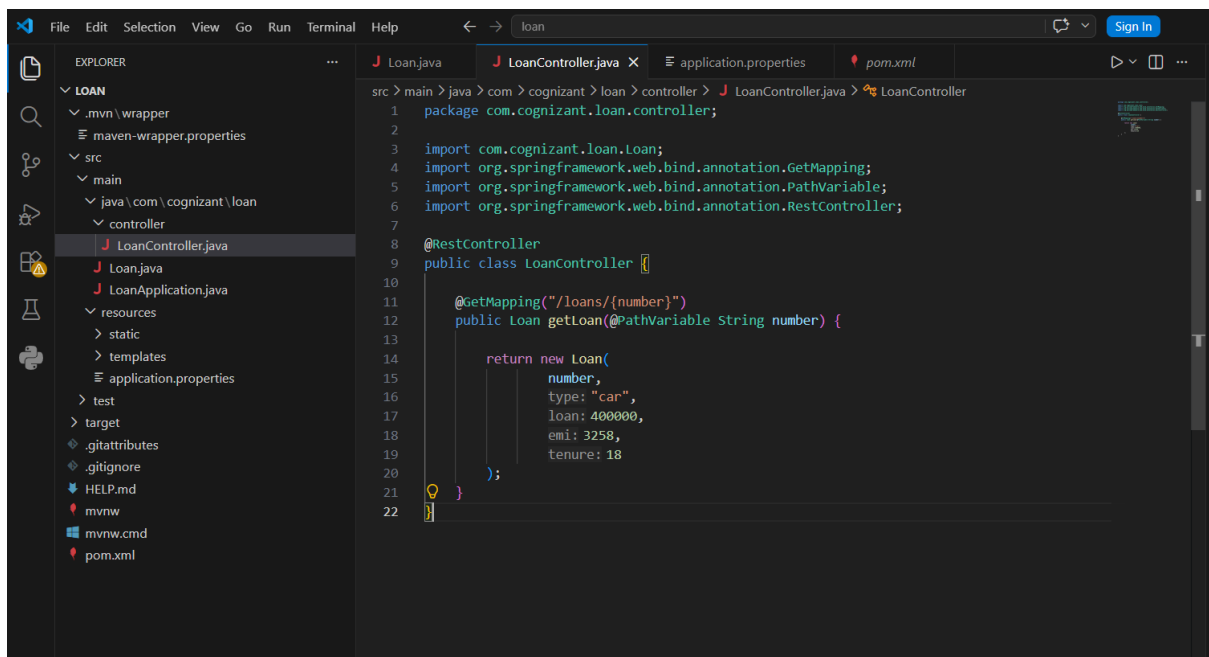
Spring Web **WEB**

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools **DEVELOPER TOOLS**

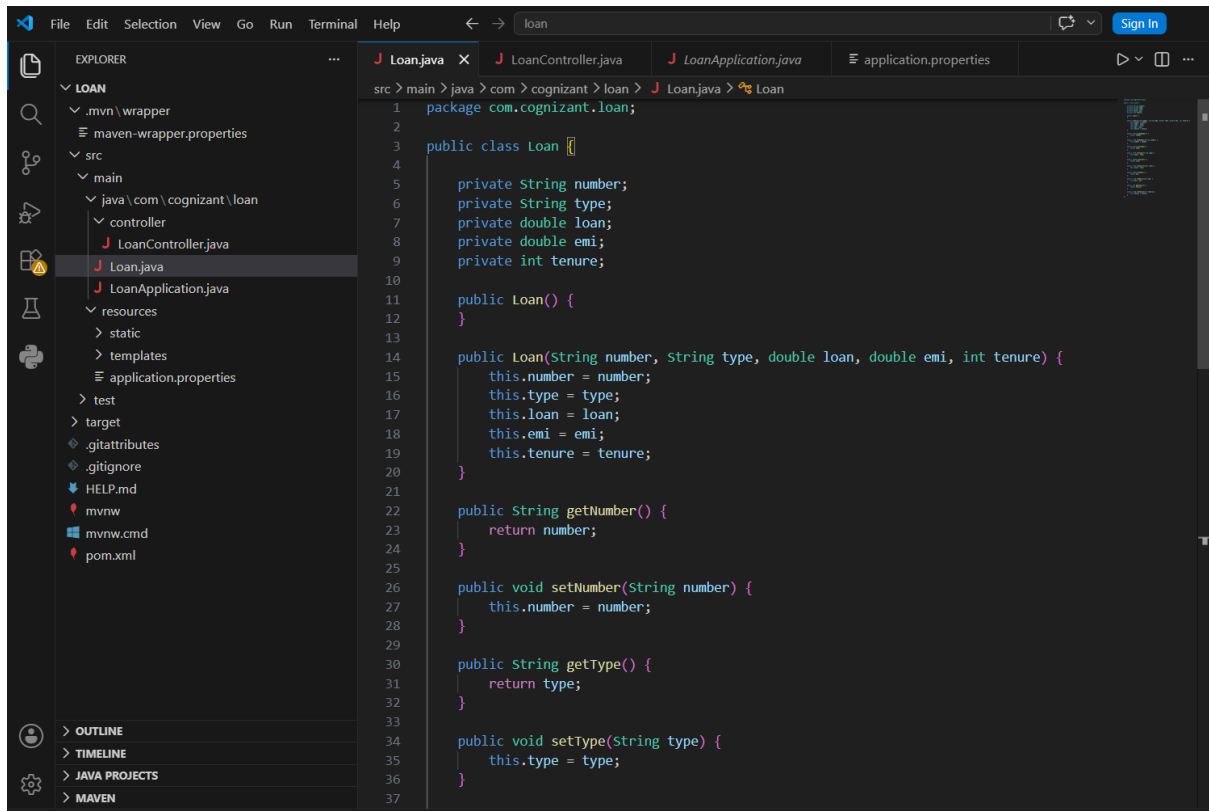
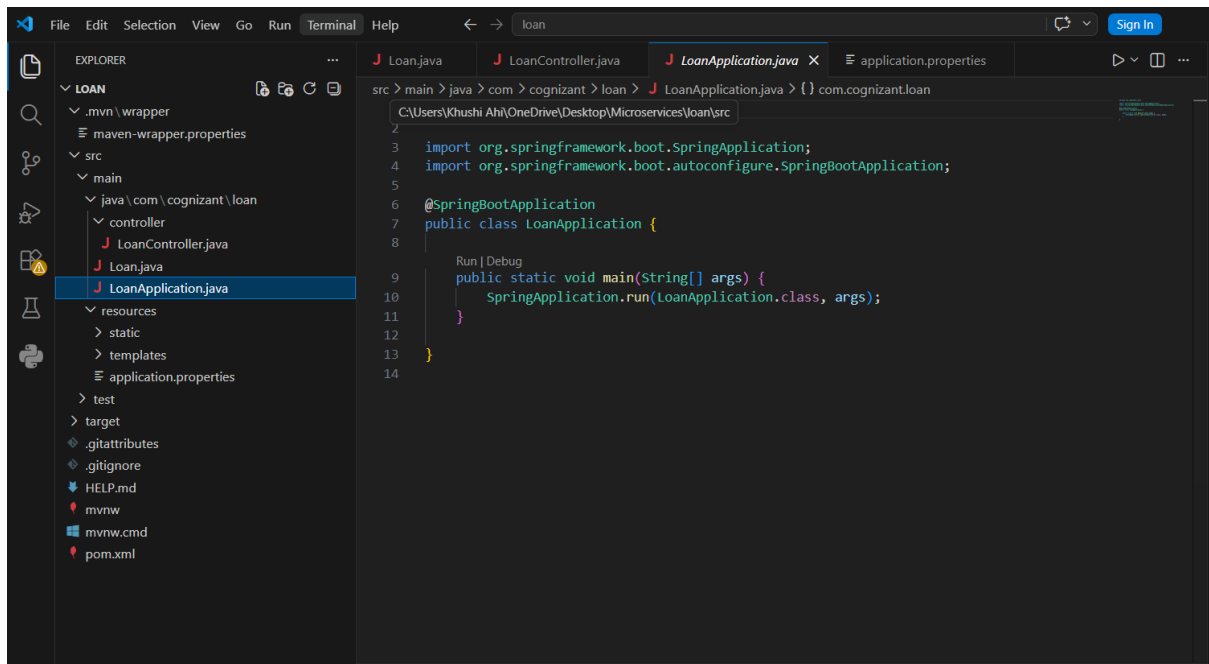
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

GENERATE CTRL + G EXPLORE CTRL + SPACE ...



The VS Code editor shows the LoanController.java file. The Explorer sidebar on the left shows the project structure: LOAN, .mvn\wrapper, maven-wrapper.properties, src, main, java\com\cognizant\loan, controller, LoanController.java, Loan.java, LoanApplication.java, resources, static, templates, application.properties, test, target, .gitattributes, .gitignore, HELP.md, mvnw, mvnw.cmd, and pom.xml. The main editor area shows the code for LoanController.java, which is a REST controller for loans.

```
src > main > java > com > cognizant > loan > controller > LoanController.java > LoanController
1 package com.cognizant.loan.controller;
2
3 import com.cognizant.loan.Loan;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.PathVariable;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class LoanController {
10
11     @GetMapping("/loans/{number}")
12     public Loan getLoan(@PathVariable String number) {
13
14         return new Loan(
15             number,
16             type: "car",
17             loan: 400000,
18             emi: 3258,
19             tenure: 18
20         );
21     }
22 }
```



```
src > main > java > com > cognizant > loan > Loan.java > Loan
3 public class Loan {
38     public double getLoan() {
39         return loan;
40     }
41
42     public void setLoan(double loan) {
43         this.loan = loan;
44     }
45
46     public double getEmi() {
47         return emi;
48     }
49
50     public void setEmi(double emi) {
51         this.emi = emi;
52     }
53
54     public int getTenure() {
55         return tenure;
56     }
57
58     public void setTenure(int tenure) {
59         this.tenure = tenure;
60     }
61 }
```

OUTPUT

```
localhost:8081/loans/H00987987972342
{"number":"H00987987972342","type":"car","loan":400000.0,"emi":3258.0,"tenure":18}
```